IE6600 Project 1
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Data Visualization Project on Work-from-Home vs. Back-to-Office Policies

Agenda:

Graph and Data Analytics

1. Overview Pattern
2. Change of Working Ways
3. Analysis by City
4. Analysis by Industry
5. Employee Satisfaction

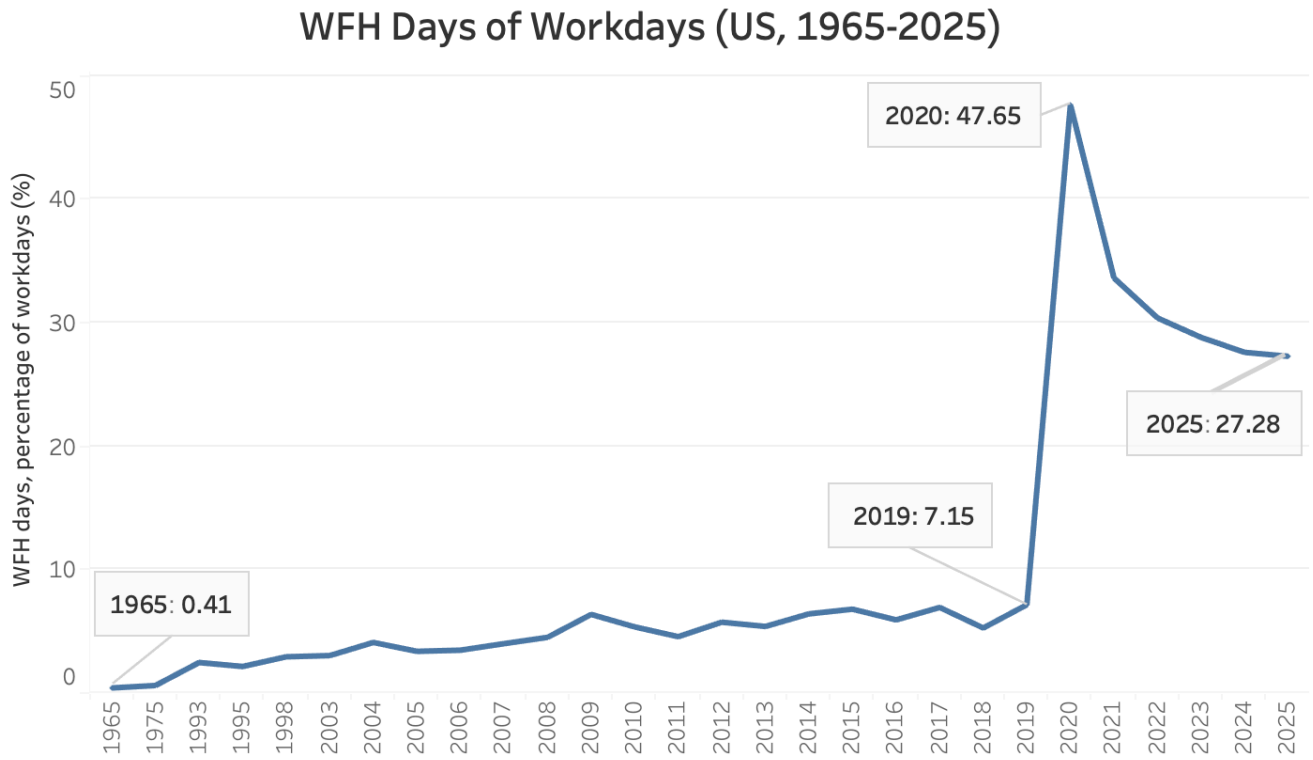
Conclusion

Data Scource

Graph and Data Analytics

1. Overview Pattern

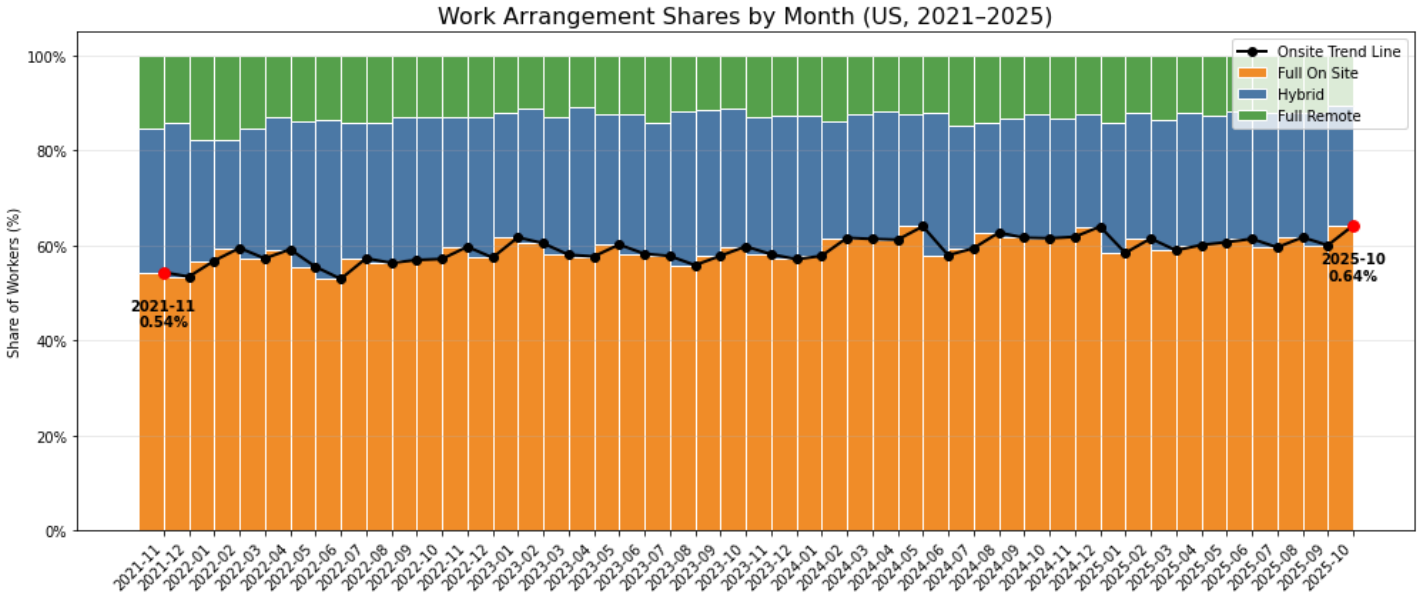
From 1965 to 2019, the share of workdays spent working from home remained very low and relatively flat. When COVID-19 hit in 2020, the share spiked to nearly 50% of all workdays. Although the percentage declined after the pandemic, it did not return to pre-COVID levels. By 2025, WFH still accounted for over 20% of workdays, indicating that remote work has become a persistent structural shift rather than a temporary shock.



WFH days as a share of all workdays in the United States, 1965–2025.
Source: Work-from-Home Research (WFHtimeseries_monthly.xlsx).

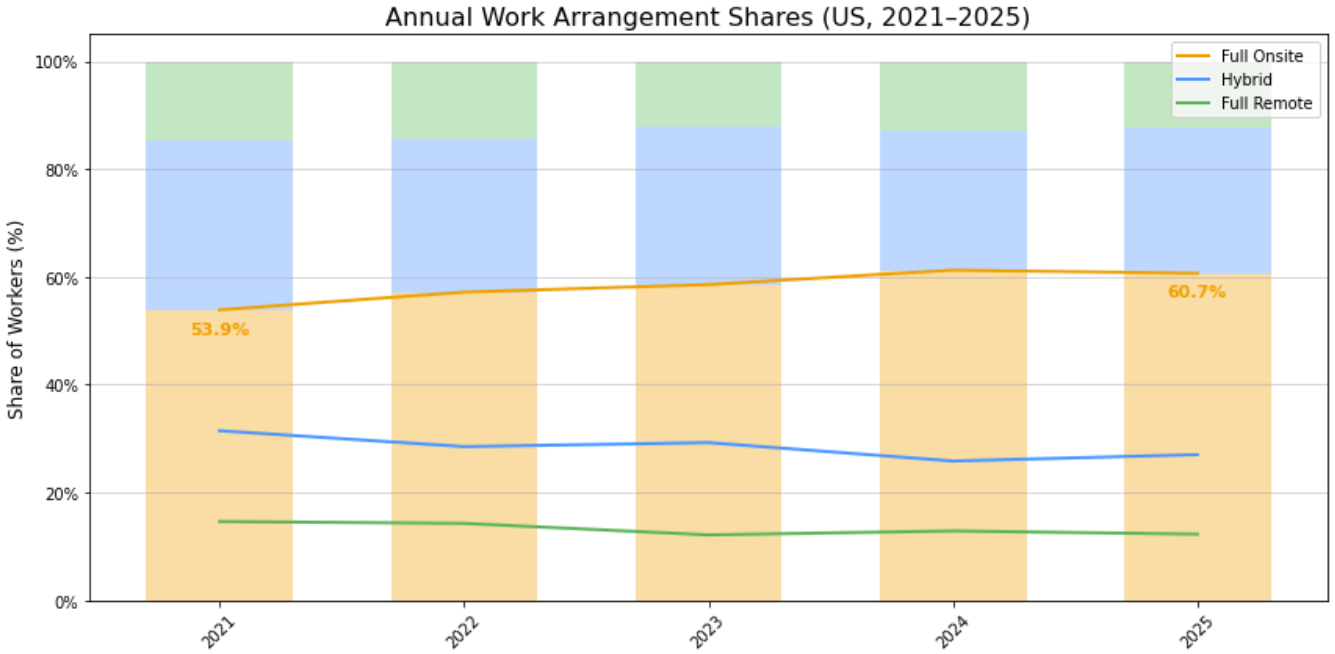
2. Change of Working Ways (Full Remote – Hybrid – Full Onsite)

Although people during the covid-19 (2021), there still had half of the working population work fully onsite. Between 2021 and 2025, the share of full-onsite workers steadily increased, while both hybrid and fully remote shares declined slightly. This indicates a gradual shift back toward office-based work after the peak of the pandemic.



Monthly shares of workers by work arrangement in the United States (Nov 2021–Oct 2025).
Source: Full Remote-Hybrid-Full Onsite (WFHtimeseries_monthly.xlsx).

The annual averages show that full-onsite work increased from about 54% in 2021 to around 61% in 2025. Hybrid and fully remote work declined slightly over the same period. This reflects a gradual rebalancing toward onsite work.



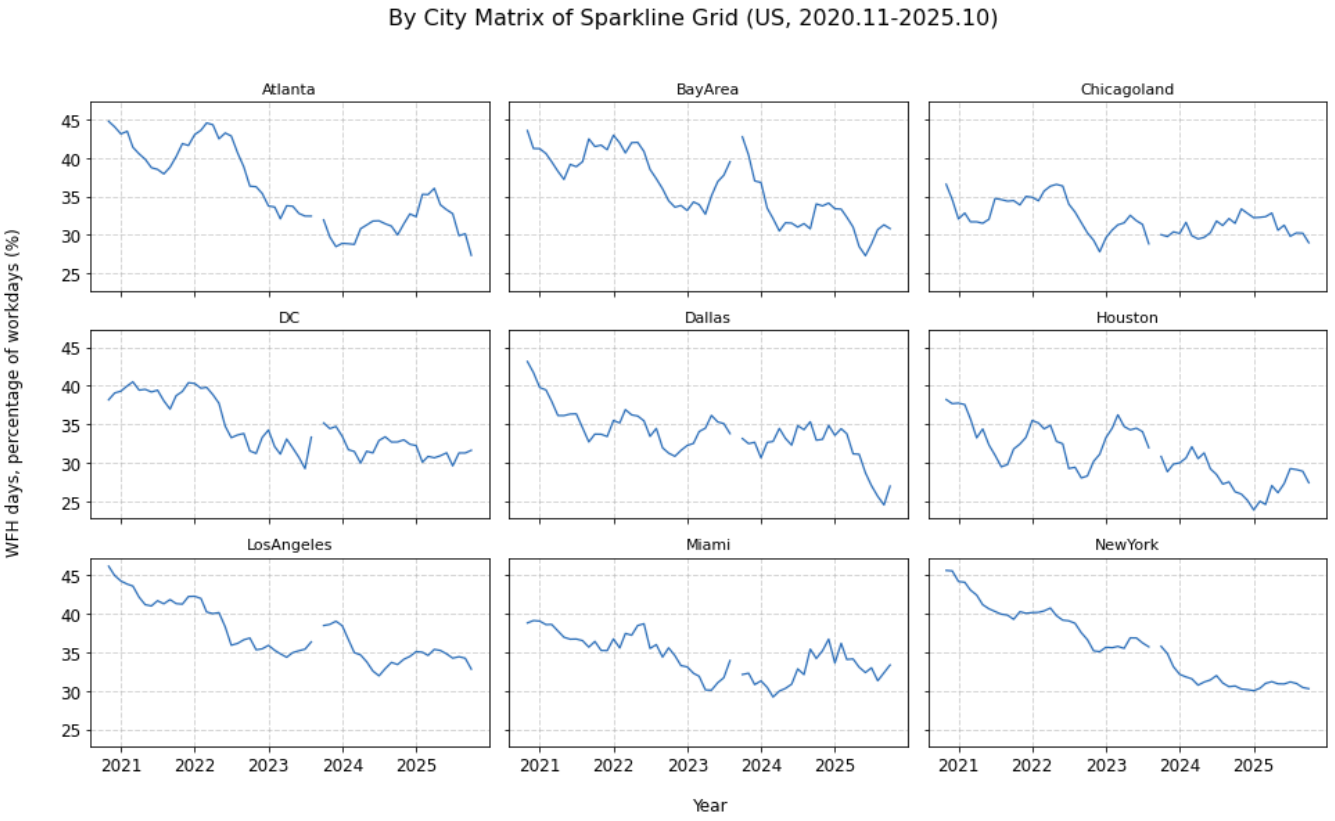
Annual average share of workers by work arrangement in the United States, 2021–2025.
Source: Full Remote-Hybrid-Full Onsite (WFHtimeseries_monthly.xlsx).

3. Analysis by City

The analysis begins in November 2020 because the WFH Research dataset applies a 6-month moving average (MA6) starting from that point. The value reported for October 2020 reflects only a shorter 3-month window and is therefore not comparable to the later months.

All WFH time-series in this section use the MA6 version provided by the WFH Research team, which applies a six-month moving average to reduce short-term volatility and highlight long-term trends.

Across the nine areas, WFH shares were relatively high around 2021 and gradually declined afterward. Cities such as the Bay Area and New York retained higher WFH levels, while places like Miami and Houston saw steeper declines.



City-level WFH shares (6-month moving average), Nov 2020–Oct 2025.
Source: WFH by city (WFHtimeseries_monthly.xlsx).
Notes about suppressed data in September 2023 due to data quality issues.

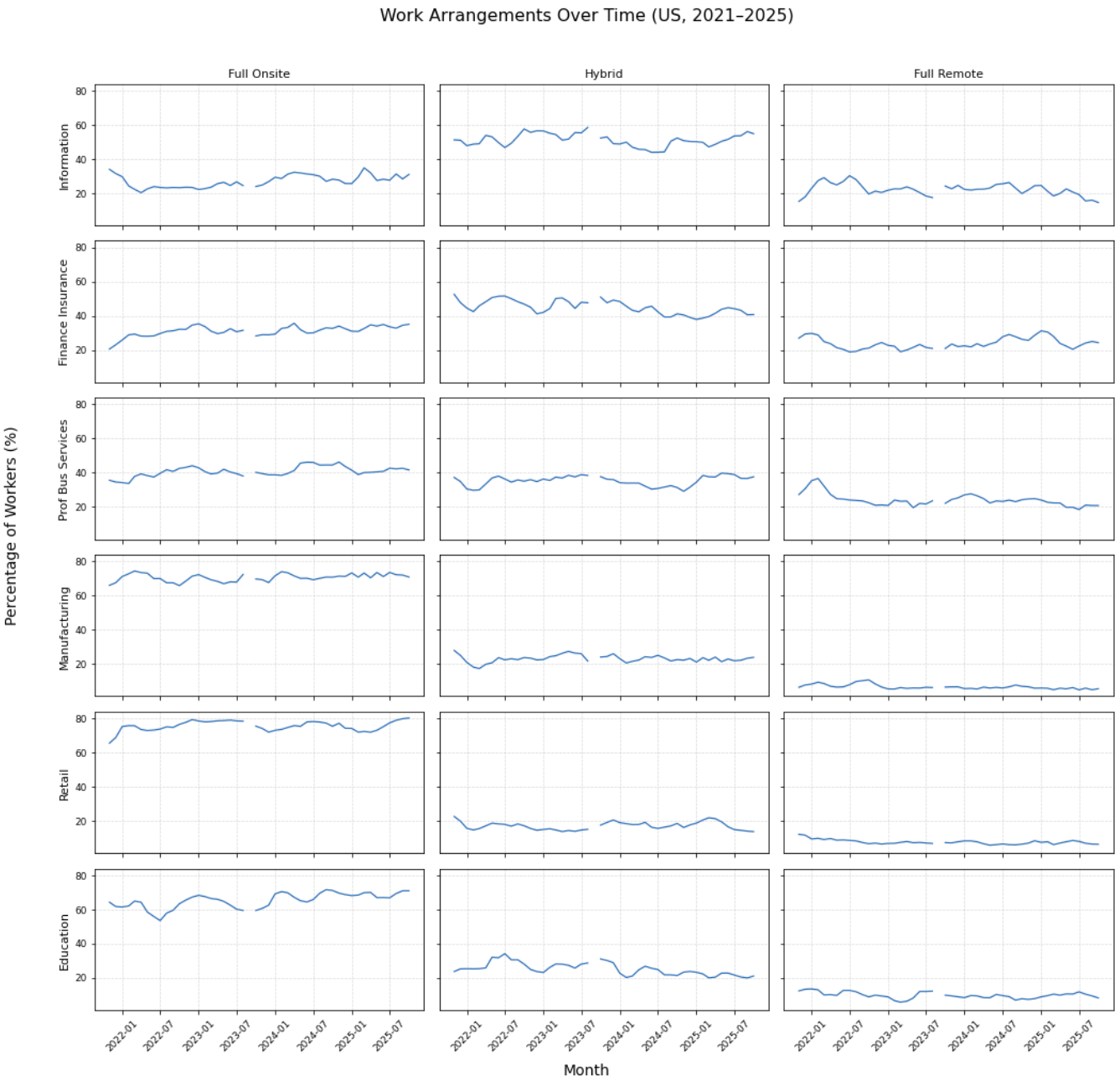
4. Analysis by Industry

There are 14 different industries in the data source. The graph shows 6 of them to analyze the path of the working conditions.

Information, Finance & Insurance, and Professional & Business Services consistently show the highest levels of hybrid and remote work. These sectors reflect jobs that can be performed digitally with minimal physical constraints.

Education briefly increased hybrid activity during the pandemic, but by 2023 it returned to onsite teaching, reflecting the sector's reliance on in-person instruction.

Manufacturing, Retail remain almost entirely onsite throughout the period, showing that those industries still remain in physical working condition.

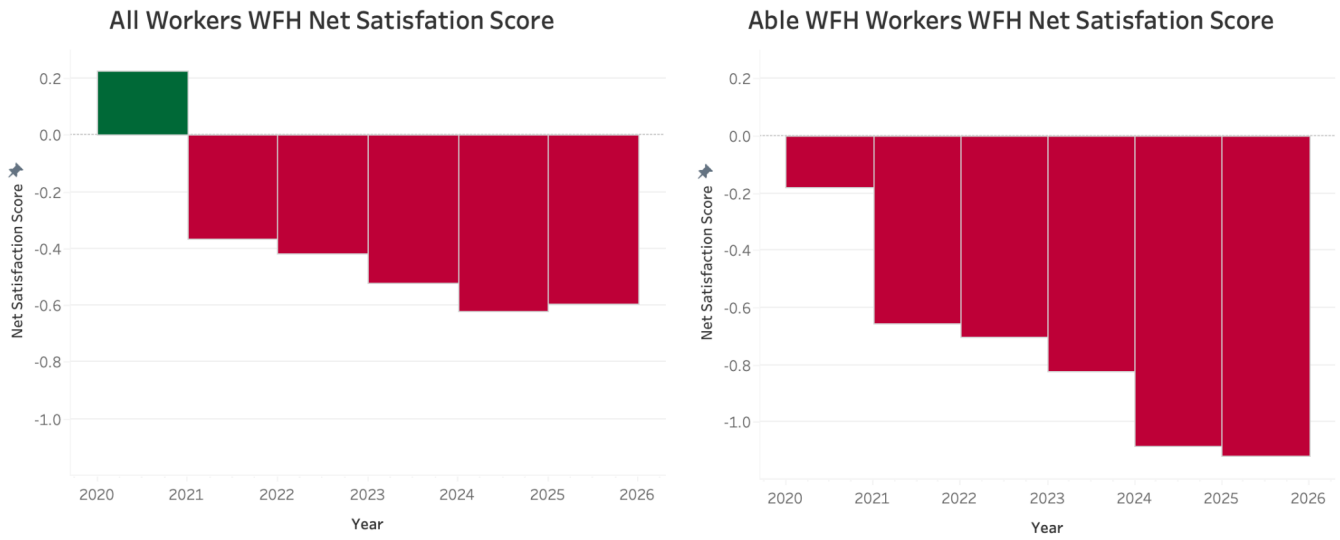


Each panel shows the share of workers in a given industry who are full onsite, hybrid, or fully remote (United States, 2021–2025).
Source: Work Arrangements by Industry (WFHtimeseries_monthly.xlsx).
Notes about suppressed data in September 2023 due to data quality issues.

5. Employee Satisfaction

Workers' preferred number of WFH days has increasingly exceeded the number of days they are actually able to work from home. For all workers, the net satisfaction score turns negative after 2021 and continues to decline through 2026, suggesting that employees generally want more flexibility than they currently receive.

This dissatisfaction is even stronger among workers whose jobs can be performed remotely. Their gap widens more quickly and reaches deeper negative values, indicating unmet expectations and growing frustration with limited WFH opportunities.



Net satisfaction gap between desired and actual work-from-home days for all workers (left) and for workers whose jobs can be performed remotely (right), United States, 2020–2026. Positive values indicate that workers received more WFH days than desired; negative values indicate they received fewer than desired (Gap = desired – actual).
Source: Worker Desires Post-COVID WFH (WFHtimeseries_monthly.xlsx).

Conclusion

The data shows that work-from-home has stabilized as a long-term feature of the labor market. Although WFH levels have declined from the 2020 peak, they remain far above pre-pandemic norms.

Cities and industries diverge sharply: tech-related and professional service sectors continue to rely heavily on hybrid or remote work, while manufacturing, retail, and other physically intensive industries remain primarily onsite. Employee satisfaction results indicate that many workers still want more flexibility than they currently receive.

Overall, the evidence supports a hybrid-first approach. Hybrid work balances flexibility with operational needs, aligns with industry differences, and better matches worker preferences, making it the most sustainable model moving forward.

Data Scource

WFH Research database: https://wfhresearch.com/